

*Non-Calculator*

**Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.**

**You have 30 minutes.**

Signature \_\_\_\_\_

**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

**1.**

a)  $-\sqrt{-49}$

b)  $6 - 2i$

Find the product. Write your answer in standard form. (2 points)

**2.**  $\sqrt{-8} \cdot \sqrt{-18}$

Simplify each power of  $i$ . (1 point each)

**3.**

a)  $i^{47}$

b)  $\frac{1}{i^{14}}$

Find the sum or difference. Write your answer in standard form. (2 points)

**4.**  $(6 - 5i) - (-2 + 3i) + (4 - i)$

Find the product. Write your answer in standard form. (2 points)

5.  $(3 - 4i)(-2 + 5i)$

Find the quotient. Write your answer in standard form. (3 points)

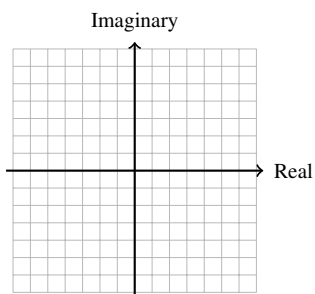
6.  $\frac{4 + 7i}{2 - 3i}$

Solve the equation over the set of complex numbers. (4 points)

7.  $2x^2 + 3 = -4x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant. (3 points)

8.  $5 - 2i$ ,  $-3 + 6i$



Write the complex number in rectangular form. (3 points)

9.  $8(\cos 300^\circ + i \sin 300^\circ)$

Write each complex number in trigonometric (polar) form, with  $\theta$  in the interval  $[0^\circ, 360^\circ)$ . (2 points each)

10.

a)  $-3 - 3i$

b)  $-5$

The Julia set: compute  $z^2 - 1$ ,  $(z^2 - 1)^2 - 1$ ,  $[(z^2 - 1)^2 - 1]^2 - 1$ , ... If the absolute value of any result exceeds 2,  $z$  is **not** in the Julia set; otherwise it is. (3 points)

11. Is  $z = i$  in the Julia set? Show work.

YES / NO

**ANSWER KEY – Non-Calculator****SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

**1.**

a)  $-\sqrt{-49}$

b)  $6 - 2i$

a)  $-\sqrt{-49} = -7i$ ;  $a = 0$ ,  $b \neq 0$ .

b)  $a \neq 0$ ,  $b = -2 \neq 0$ .

complex, pure imaginary, nonreal complex

complex, nonreal complex

Find the product. Write your answer in standard form. (2 points)

**2.**  $\sqrt{-8} \cdot \sqrt{-18}$

Pull out  $i$  first:  $\sqrt{-8} = 2i\sqrt{2}$ ,  $\sqrt{-18} = 3i\sqrt{2}$ 

$(2i\sqrt{2})(3i\sqrt{2}) = 6i^2(2) = 12i^2 = \boxed{-12}$  ( $\sqrt{144} = 12$  is the trap: the sign flips.)

Simplify each power of  $i$ . (1 point each)

**3.**

a)  $i^{47}$

b)  $\frac{1}{i^{14}}$

a)  $47 \div 4 = 11 \text{ R } 3$ , so  $i^{47} = i^3 = \boxed{-i}$

b)  $14 \div 4 = 3 \text{ R } 2$ , so  $i^{14} = i^2 = -1$ ;  $\frac{1}{-1} = \boxed{-1}$

Find the sum or difference. Write your answer in standard form. (2 points)

**4.**  $(6 - 5i) - (-2 + 3i) + (4 - i)$

$6 - 5i + 2 - 3i + 4 - i$  real:  $6 + 2 + 4 = 12$  imaginary:  $-5 - 3 - 1 = -9$

$\boxed{12 - 9i}$

Find the product. Write your answer in standard form. (2 points)

**5.**  $(3 - 4i)(-2 + 5i)$

$= -6 + 15i + 8i - 20i^2 = -6 + 23i + 20 = \boxed{14 + 23i}$

Find the quotient. Write your answer in standard form. (3 points)

**6.**  $\frac{4 + 7i}{2 - 3i}$

Multiply by the conjugate of the bottom:  $\frac{4 + 7i}{2 - 3i} \cdot \frac{2 + 3i}{2 + 3i}$ 

top:  $8 + 12i + 14i + 21i^2 = -13 + 26i$  bottom:  $4 + 9 = 13$

$\frac{-13 + 26i}{13} = \boxed{-1 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7.  $2x^2 + 3 = -4x$

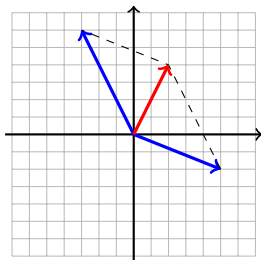
$2x^2 + 4x + 3 = 0$ ,  $a = 2$ ,  $b = 4$ ,  $c = 3$        $b^2 - 4ac = 16 - 24 = -8$

$$x = \frac{-4 \pm \sqrt{-8}}{4} = \frac{-4 \pm 2i\sqrt{2}}{4} = \boxed{-1 \pm \frac{\sqrt{2}}{2}i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant. (3 points)

8.  $5 - 2i$ ,  $-3 + 6i$

$$(5 - 3) + (-2 + 6)i = \boxed{2 + 4i}$$



Resultant (red) is the diagonal of the parallelogram.

Write the complex number in rectangular form. (3 points)

9.  $8(\cos 300^\circ + i \sin 300^\circ)$

$300^\circ$ : QIV, reference  $60^\circ$ ;  $\cos 300^\circ = \frac{1}{2}$ ,  $\sin 300^\circ = -\frac{\sqrt{3}}{2}$

$$8\left(\frac{1}{2}\right) + 8\left(-\frac{\sqrt{3}}{2}\right)i = \boxed{4 - 4i\sqrt{3}}$$

Write each complex number in trigonometric (polar) form, with  $\theta$  in the interval  $[0^\circ, 360^\circ)$ . (2 points each)

10.

a)  $-3 - 3i$

b)  $-5$

a) QIII.  $r = \sqrt{9+9} = 3\sqrt{2}$

$\tan \theta = \frac{-3}{-3} = 1$ , ref  $45^\circ$

$\theta = 180^\circ + 45^\circ = 225^\circ$

$$\boxed{3\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)}$$

b) negative real axis

$r = 5$ ,  $\theta = 180^\circ$

$$\boxed{5(\cos 180^\circ + i \sin 180^\circ)}$$

The Julia set: compute  $z^2 - 1$ ,  $(z^2 - 1)^2 - 1$ ,  $[(z^2 - 1)^2 - 1]^2 - 1$ , ... If the absolute value of any result exceeds 2,  $z$  is **not** in the Julia set; otherwise it is. (3 points)

11. Is  $z = i$  in the Julia set? Show work.

$i^2 - 1 = -2$ ,  $|-2| = 2$ : exactly 2 does not exceed 2, keep going.

$(-2)^2 - 1 = 3$ ,  $|3| = 3 > 2$ .

**NO**